PRODUCT REQUIREMENTS DOCUMENT (PRD)

Carbon Horizon

Learn Your Impact. Shape the Future.

Version: 2.0

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1. Executive Summary

Carbon Horizon is an AI-powered Personal Sustainability Intelligence Platform that helps individuals measure, simulate, forecast, and reduce their carbon footprint through actionable recommendations, behavioral tracking, and future impact forecasting.

Unlike traditional carbon footprint calculators that only provide emission estimates, Carbon Horizon focuses on helping users understand which actions matter most and guides them toward measurable environmental improvement.

The platform combines:

- Carbon Accounting
- Sustainability Intelligence
- AI-Powered Guidance
- Carbon Reduction Simulation
- Lifestyle Impact Forecasting
- Sustainability Goal Tracking

into a single unified experience.

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2. Problem Statement

Millions of people want to live sustainably but struggle to answer critical questions:

- What is causing most of my carbon emissions?
- Which changes will have the biggest impact?
- How much carbon can I realistically reduce?
- How can I maintain sustainable habits over time?
- How can I measure progress?

Existing carbon calculators provide static reports but rarely help users take action.

Carbon Horizon transforms awareness into measurable carbon reduction.

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3. Product Vision

To become the personal sustainability intelligence platform that empowers people to make informed environmental decisions and continuously improve their carbon footprint through data-driven insights.

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4. Product Objectives

Business Objectives

- Build a competition-ready MVP
- Demonstrate measurable sustainability impact
- Showcase AI-assisted environmental decision making
- Provide a scalable foundation for future growth

User Objectives

- Understand environmental impact
- Discover major emission sources
- Simulate alternative choices
- Receive personalized recommendations
- Set reduction goals
- Track sustainability progress

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5. Success Metrics (KPIs)

Primary KPIs

- Assessment Completion Rate > 80%
- Dashboard Engagement Rate > 60%
- Simulation Usage Rate > 50%
- Goal Creation Rate > 40%
- Habit Tracking Retention (7 Days) > 30%
- AI Coach Usage Rate > 50%

Environmental KPIs

- Estimated Carbon Reduction Generated
- Total Carbon Saved by Users

- Reduction Goal Achievement Rate
- Habit Completion Rate

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6. Target Users

Students

Needs:

- Environmental awareness
- Low-cost sustainability actions
- Transportation optimization

Pain Points:

- Limited budget
- Lack of environmental knowledge

Working Professionals

Needs:

- Commute optimization
- Energy reduction
- Lifestyle improvements

Pain Points:

- Limited time
- Difficulty identifying impactful actions

Families

Needs:

- Household sustainability planning
- Energy management
- Waste reduction

Pain Points:

- Multiple contributors to emissions

- Tracking household habits

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7. User Personas

Persona 1 – Environmentally Conscious Student

Age: 20

Goals:

- Reduce carbon footprint
- Learn sustainable practices

Challenges:

- Limited budget
- Limited environmental knowledge

Persona 2 – Working Professional

Age: 30

Goals:

- Reduce commuting emissions
- Improve energy efficiency

Challenges:

- Busy schedule
- Complex sustainability information

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8. User Stories

Assessment

As a user,

I want to calculate my carbon footprint,

So that I understand my environmental impact.

As a user,

I want to see my largest emission source,

So that I know where to focus improvement efforts.

AI Coach

As a user,

I want personalized recommendations,

So that I know which actions are most effective.

Simulator

As a user,

I want to test lifestyle changes,

So that I can understand their impact before implementing them.

Forecast

As a user,

I want to visualize future emissions,

So that I can plan long-term improvements.

Goals

As a user,

I want to set carbon reduction goals,

So that I can track progress toward sustainability targets.

Habit Tracking

As a user,

I want to track daily eco-friendly activities,

So that I can build sustainable habits.

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9. Core Product Features

Feature 1: Carbon Footprint Assessment

Purpose

Calculate user carbon emissions.

Inputs

Transportation

- Car
- Motorcycle
- Bus
- Train
- Flight
- Bicycle

Energy

- Monthly Electricity Usage
- AC Usage
- LPG/Gas Usage
- Solar Usage

Food

- Vegetarian
- Mixed
- Non-Vegetarian

Waste

- Recycling Frequency
- Plastic Usage

Household

- Household Members

Outputs

- Daily Emissions
- Monthly Emissions
- Annual Emissions
- Emission Breakdown
- Carbon Score

Feature 2: Carbon Intelligence Dashboard

Displays:

- Annual Carbon Footprint
- Carbon Score
- Reduction Potential
- Top Emission Source
- Trend Analysis
- Goal Progress

Visualizations:

- Donut Charts
- Line Charts
- Progress Indicators
- Goal Completion Graphs

Feature 3: AI Sustainability Coach

Purpose:

Provide personalized sustainability guidance.

Important Rule:

AI never performs carbon calculations.

All calculations are generated by the backend calculation engine.

AI only:

- Explains results
- Provides recommendations
- Generates action plans
- Answers sustainability questions

Feature 4: Carbon Reduction Simulator

Purpose:

Allow users to test future lifestyle changes.

Supported Simulations:

Transportation

- Car → Bus
- Motorcycle → Train
- Car → Bicycle

Energy

- Electricity Reduction
- Reduced AC Usage
- Solar Adoption

Food

- Non-Veg → Mixed
- Mixed → Vegetarian

Waste

- Improved Recycling

Outputs:

- Carbon Saved
- Reduction Percentage
- Future Impact
- Estimated Annual Savings

Feature 5: Lifestyle Impact Forecast

Purpose:

Project future emissions based on:

- Current Emissions
- User Habits
- Goal Progress
- Simulated Changes

Scenarios:

Scenario A

Current Lifestyle Continues

Scenario B

Partial Improvements

Scenario C

Recommended Improvements

Outputs:

- 3-Month Projection
- 6-Month Projection
- 12-Month Projection

Feature 6: Sustainability Goal Management

Users can:

- Create Goals
- Track Progress
- Receive Goal-Based Recommendations
- Monitor Completion

Example:

Goal: Reduce annual emissions by 20%

Target Date: 31 Dec 2026

Feature 7: Eco Habit Tracker

Track:

- Public Transport Usage
- Electricity Savings
- Recycling
- Water Conservation
- Plastic Reduction

Outputs:

- Streaks
- Weekly Reports
- Monthly Reports
- Carbon Saved

Feature 8: Assessment History

Users can:

- View previous assessments
- Compare progress over time
- Recalculate footprint

Feature 9: Simulation History

Users can:

- Save simulations
- Compare scenarios
- Reuse previous simulations

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10. Carbon Methodology Requirements

The platform shall use scientifically recognized emission factors.

Sources may include:

- IPCC

- EPA
- IEA
- Government Energy Statistics

Method:

Carbon Emission = Activity Data × Emission Factor

All emission factors must be versioned and traceable.

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11. Carbon Score System

Purpose:

Provide a simple sustainability indicator.

Formula (MVP):

Carbon Score = 100 − Normalized Emission Impact

Score Range:

- 80–100 → Excellent
- 60–79 → Good
- 40–59 → Moderate
- Below 40 → Needs Improvement

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12. Functional Requirements

FR-1 User Authentication

FR-2 Carbon Assessment

FR-3 Carbon Score Generation

FR-4 Dashboard Generation

FR-5 AI Recommendation Generation

FR-6 Simulation Engine

FR-7 Forecast Generation

FR-8 Habit Tracking

FR-9 Goal Management

FR-10 Assessment History

FR-11 Simulation History

FR-12 Data Persistence

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13. Acceptance Criteria

Carbon Assessment

Given valid user inputs

When user submits assessment

Then annual emissions shall be generated within 2 seconds.

Dashboard

Given completed assessment

When dashboard loads

Then breakdown charts shall display correctly.

Simulator

Given selected lifestyle changes

When simulation runs

Then projected emissions shall be calculated accurately.

Forecast

Given historical habits and emissions

When forecast is generated

Then system shall display 3, 6, and 12 month projections.

AI Coach

Given assessment results

When user asks a question

Then AI shall generate a personalized response within 5 seconds.

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14. Non-Functional Requirements

Performance

- API Response < 2 Seconds
- Dashboard Load < 3 Seconds

Scalability

- 1,000+ Users

Availability

- 99% Uptime

Accessibility

- WCAG AA Compliance

Security

- JWT Authentication
- bcrypt Password Hashing
- Input Validation
- Rate Limiting
- Secure Secret Storage

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15. User Retention Strategy

MVP Retention Features

- Goal Tracking
- Weekly Sustainability Reports
- Progress Milestones
- Habit Streaks
- Monthly Carbon Summary

Future Versions

- Badges
- Leaderboards
- Community Challenges

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16. Future Scope (Version 2.0+)

- OCR Bill Scanner
- Mobile Application
- Community Challenges
- Smart Device Integration
- Gamification
- Carbon Budget Planner
- Carbon Marketplace
- Machine Learning Forecasting

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17. Technology Stack

Frontend

- React + TypeScript
- Tailwind CSS
- Chart.js

Backend

- FastAPI

Database

- PostgreSQL

AI

- Gemini API

Authentication

- JWT

Deployment

- Docker
- Google Cloud Run

CI/CD

- GitHub Actions

Monitoring

- Google Cloud Logging

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18. Product Success Criteria

The product shall be considered successful when:

- Users can accurately calculate emissions
- Users understand major emission contributors
- AI provides actionable recommendations
- Simulations demonstrate measurable reductions
- Forecasts visualize future impact
- Users actively track sustainability goals
- The application is successfully deployed and demonstrated during evaluation